Fiscal Year:	FY 2008	**Task Last Updated:**	FY 08/27/2008
PI Name:	Baldwin, Kenneth M. Ph.D.		
Project Title:	Force Regulation As a Countermeasure To Muscle Atrophy		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Muscle Alterations and Atrophy Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Irvine	**State:**	CA
Zip Code:	92697-4560	**Congressional District:**	48
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	03/01/2004	**End Date:**	04/30/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	3
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	5
No. of Bachelor's Candidates:	8	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 4/30/2008 per K. Major/NSBRI (3/08)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Adams, Gregory (University of California, Irvine) Caiozzo, Vincent (University of California, Irvine) Haddad, Fadia (University of California, Irvine)		
Grant/Contract No.:	NCC 9-58-MA00402		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	In recent years considerable attention has focused on identifying exercise paradigms to ameliorate the deficits of atrophy and reduced strength in astronaut muscle properties, especially given that exposure to spaceflight is now minimally six months or longer on the International Space Station (ISS). While there is evidence that exercise of the resistance type is partially successful in reducing the muscle atrophy response, a definitive exercise prescription has not been identified that is suitable for conditioning skeletal muscle given the various constraints of microgravity that impact the homeostasis of astronauts. The primary goals of the proposal are to utilize a rodent resistance training model in order to: 1) ascertain the efficacy of acute bouts of isometric, concentric, eccentric, and concentric-eccentric modes of contraction, as performed under conditions of muscle unloading, in their ability to activate molecular markers indicative of anabolic responses that have been shown to favor a positive protein balance profile in order to blunt the atrophy process; 2) test and optimize the most promising paradigm(s) for its efficacy in ameliorating the atrophy of muscle associated with the model of hindlimb suspension; and 3) determine the efficacy of using resistance training, in combination with other pharmacological interventions (such as protease inhibiters), as to their ability to ameliorate muscle atrophy-induced processes as compared to resistance training alone. Ultimately, the objective is to identify a resistance training prescription that can be efficiently performed (minimizing time constraints), requires simple, yet reliable equipment devices, and is attractive to the astronauts for compliance.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our project provides insight on practical ways to improve muscle mass and lean body mass in both young and old individuals. It has application to sports training and a means to helping aging individuals maintain strength in their limb muscles thus reducing accidents and injury. Furthermore, our findings indicate that sophisticated exercise modalities are not necessary for inducing muscle mass and strength. Extending this approach by using modes of exercise that can train the muscle for both endurance and strength will allow for a better prescription for conserving the homeostasis of skeletal muscle in the face on unloading stimuli.
Task Progress:	This project serves as a foundation study in which we will build out to address the four specific aims of the project. Based on the findings in the 2006-2007 funding period we have begun to limit the variables necessary to control in establishing a resistance exercise prescription in the face of muscle atrophy states. This would involve adding more volume of exercise per training session regardless of the mode that is being used to activate the muscle as long as sufficient loading is imposed on the muscle during each contraction. In addition we have found that 7 contractions to failure spaced over 4 bouts may be the optimal stimulus during each training session.
Bibliography Type:	Description: (Last Updated: 04/21/2011)
Articles in Peer-reviewed Journals	Adams GR, Haddad F, Bodell PW, Tran PD, Baldwin KM. "Combined isometric, concentric, and eccentric resistance exercise prevents unloading-induced muscle atrophy in rats." J Appl Physiol. 2007 Nov;103(5):1644-54. Epub 2007 Sep 13. PMID: 17872405 , Nov-2007